Hints: Inequalities – 3. Integral inequalities: the working techniques

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Let $M = \max_{[0,1]} |f'|$ and $F(x) = \int_0^x f$. Compare $f(x)^2$ with $f(0)^2$ through the derivative of f^2.

Hint 2. $f(x)^2 - f(0)^2 = \int_0^x 2ff' \leqslant 2M \int_0^x f = 2MF(x)$. Multiply by $f(x) \geqslant 0$: $f^3 - f(0)^2 f \leqslant 2MfF = M(F^2)'$.

Hint 3. Integrate over $[0,1]$: $\int_0^1 f^3 - f(0)^2 \int_0^1 f \leqslant MF(1)^2 = M\big(\int_0^1 f\big)^2$.

Problem 2

Hint 1. Recognise $2f' + xf''$ as a second derivative of a simple function built from f.

Hint 2. $h(x) = xf(x)$ satisfies $h'' = 2f' + xf'' \geqslant 1$ and $h(0) = 0$. So $h(x) - \frac{x^2}{2}$ is convex.

Hint 3. A convex function lies above its tangent at 0: $h(x) - \frac{x^2}{2} \geqslant h'(0)x$. Integrate over $(-1,1)$; the linear term integrates to 0, and $\int_{-1}^1 \frac{x^2}{2} = \frac{1}{3}$.

Problem 3

Hint 1. Put $u = \sqrt{f}$, $v = \sqrt{g}$ (non-decreasing) and $\varphi(t) = \sqrt{1+t^2}$, which is convex. This is an integral form of the majorisation (Karamata) inequality.

Hint 2. By convexity, $\varphi(u) - \varphi(v) \geqslant \varphi'(v)(u - v)$, and $\varphi'(v(x))$ is non-decreasing in x. Let $H(x) = \int_a^x (u - v) \leqslant 0$, $H(a) = H(b) = 0$.

Hint 3. Integrate by parts (Stieltjes, or approximate v by smooth functions): $\int_a^b \varphi'(v)\, dH = -\int_a^b H\, d\varphi'(v) \geqslant 0$, since $H \leqslant 0$ and $\varphi'(v)$ is non-decreasing.

Problem 4

Hint 1. Split $[0,1]$ into $P = \{f \geqslant 0\}$ and $N = \{f < 0\}$, with measures p, n ($p + n = 1$), and put $A = \int_P f$, $B = \int_N |f|$. On $P \times P$ and $N \times N$ the integrand is $|f(x)| + |f(y)|$.

Hint 2. So $\iint |f(x) + f(y)| \geqslant 2pA + 2nB + 2\big|\iint_{P\times N}(f(x) + f(y))\big| = 2pA + 2nB + 2|nA - pB|$, and you need this to be $\geqslant A + B$.

Hint 3. By symmetry assume $nA \geqslant pB$. The bound becomes $2A + 2B(n-p)$, and $2A + 2B(n-p) \geqslant A + B$ reduces to $A \geqslant (1 - 2n + 2p)B$; it follows from $A \geqslant \frac{p}{n}B$ since $\frac{p}{n} \geqslant 1 - 2n + 2p \iff 1 \geqslant 4pn$.

Problem 5

Hint 1. The polynomials of degree at most 1999 form a finite-dimensional vector space.

Hint 2. $\|p\| = \int_{-1}^1 |p|$ is a norm on it (a non-zero polynomial vanishes only at finitely many points) and $p \mapsto |p(0)|$ is continuous. On the compact unit sphere $\{\|p\| = 1\}$ the latter is bounded by some C; scale.

Problem 6

Hint 1. Integrate $\int_0^1 f$ by parts to make f' appear, and use $f(1) = -\frac{1}{6}$.

Hint 2. $\int_0^1 f = f(1) - \int_0^1 xf' = -\frac{1}{6} - \int_0^1 xf'$. The condition becomes $\int_0^1 ((f')^2 + 2xf') + \frac{1}{3} \leqslant 0$, and $\frac{1}{3} = \int_0^1 x^2$.

Hint 3. So $\int_0^1 (f' + x)^2 \leqslant 0$, hence $f'(x) = -x$ and $f(x) = \frac{1}{3} - \frac{x^2}{2}$ is the only solution.

Problem 7

Hint 1. f is an increasing bijection of $[0, 1]$ with $f(1) = 1$. Interpret both integrals as double integrals over parts of the unit square cut by the graph of f.

Hint 2. $\int_0^1 x^\alpha f(x)\, dx = \iint_{y \leqslant f(x)} x^\alpha$, and, by the reflection $(x, y) \mapsto (y, x)$ which maps the graph of f^{-1} to that of f, $\int_0^1 x^\alpha f^{-1}(x)\, dx = \iint_{y \geqslant f(x)} y^\alpha$.

Hint 3. Each point of the square contributes x^α or y^α, hence at most $\max(x, y)^\alpha$, and $\iint \max(x, y)^\alpha = 2\int_0^1 x^{\alpha+1}\, dx = \frac{2}{\alpha+2}$.

Problem 8

(a)

Hint 1. $g(x) = \big((x-1)f(x)\big)'$, so $\int_0^1 g = \big[(x-1)f\big]_0^1 = 0$.

(b)

Hint 1. Let $G(u) = \int_0^u g = (u-1)f(u)$. Since f is non-decreasing with $f(0) = 0$, $f \geqslant 0$ and $G \leqslant 0$, $G(0) = G(1) = 0$.

Hint 2. A convex φ with $\varphi(0) = 0$, $\varphi(1) = 1$, values in $[0, 1]$, is non-decreasing; it may vanish on some $[0, t_0]$ and is then strictly increasing with a concave inverse ψ on $[0, 1]$. Change variables $u = \varphi(t)$: $\int_0^1 g(\varphi(t))\, dt = t_0\, g(0) + \int_0^1 g(u)\psi'(u)\, du$, with $g(0) = -f'(0) \leqslant 0$.

Hint 3. The weight $\psi' \geqslant 0$ is non-increasing. Integrate by parts (Abel): $\int_0^1 g\,\psi' = \big[G\psi'\big]_0^1 - \int_0^1 G\, d\psi'$, where the boundary term vanishes and $-\int G\, d\psi' \leqslant 0$ because $G \leqslant 0$ and $d\psi' \leqslant 0$. (Where ψ' is unbounded near 0, cut off at ε and let $\varepsilon \to 0$.)

Problem 9

Hint 1. $h(t) = t + \sin t$ is an increasing bijection of $\mathbb{R}$ (with $h' \geqslant 0$ vanishing only at isolated points), so the hypothesis says $f(x) \geqslant h^{-1}(x)$.

Hint 2. h maps $[0, \pi]$ onto $[0, \pi]$, and the integrals of a function and its inverse fill a rectangle: $\int_0^\pi h^{-1} = \pi^2 - \int_0^\pi (t + \sin t)\, dt = \pi^2 - \frac{\pi^2}{2} - 2$.

Problem 10

Hint 1. Let $\Phi(c) = \frac{1}{c-a}\int_a^c f$, the mean of f on $[a, c]$. Since f is strictly increasing, $\Phi(c) < f(c)$.

Hint 2. $\Phi'(c) = \frac{f(c) - \Phi(c)}{c-a} > 0$, so Φ is strictly increasing and $\Phi(c) < \Phi(b)$. One-line version: the mean of f over $[a, c]$ is $< f(c) <$ the mean over $[c, b]$, and the mean over $[a, b]$ is a weighted average of these two.

Problem 11

Hint 1. Interpret $\sqrt{1 + \cos^2 y}$ as the arc-length element of a familiar curve, and $\sqrt{x^2 + \sin^2 x}$ as a distance.

Hint 2. The left side is the length of the graph of $y = \sin t$ for $t \in [0, 1]$; the right side is the length of the chord from $(0, 0)$ to $(x, \sin x)$. The chord is at most the arc over $[0, x]$, which is strictly shorter than the arc over $[0, 1]$ since $x < 1$.